



Identification of Canine Mast Cell Progenitors in Blood



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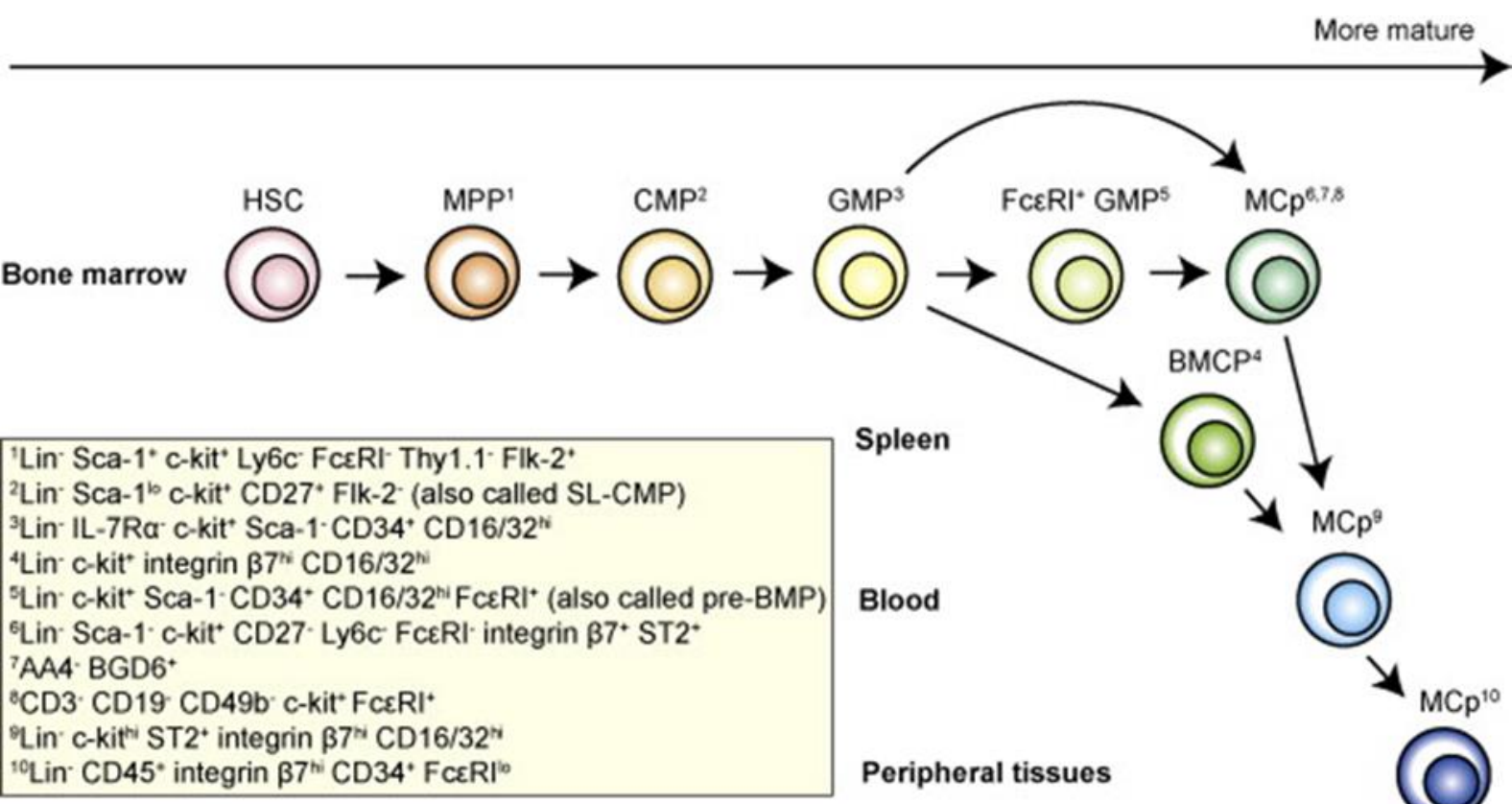
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Abstract

Mast cells are widely considered a multi-functional master cell, with involvement in histamine allergic reactions, wound healing, tissue remodeling and innate immunity. They are strongly associated with the tumor micro environment as well as pathologies such as anaphylaxis, allergic rhinitis and infantile asthma as well as a myriad of other pathologies. Mast cells are found in nearly every tissue of the body, with the exception of the central nervous system and the retina of the eye, but originate in the bone marrow as hematopoietic stem cells. These early mast cells leave the marrow and circulate in the blood as uncommitted mast cell progenitors (MCp) which are CD117⁺, CD34⁺, FcεRI⁺ and CD90⁻ before being recruited to the peripheral tissues where they mature. When canine mast cells become neoplastic they, most commonly, form tumors in the skin, but may also cause tumors in the liver, intestines, spleen and elsewhere. Our long-term goal is to sequence mast cell tumors in a patient-specific, precision medicine approach designed to identify therapeutic targets in individual patients. Normal mast cells from canine mast cell tumor patients are needed to compare transcriptomes to patient mast cell tumors using deep sequencing/single cell sequencing to aid in the identification of therapeutic targets. The current project seeks to identify and obtain appropriate normal mast cell precursors from patient animals with minimal additional morbidity. Circulating MCp's represent a population of approximately 0.1-0.5% nucleated cells in the blood, making them a difficult target to isolate. We have developed a method of Flow cytometric analysis and sorting of CD117⁺, CD34⁺, FcεRI⁺ and CD90⁻ MCp's from canine blood using antibody labeling. Once isolated by cell sorting, these cells will be harvested, their RNA sequenced, and then the RNA sequencing data compared to that of tumor cells from the same patient.

Introduction

Mast cells begin their lives in the bone marrow as hematopoietic stem cells and lose their ability for self-renewal upon development into multipotent progenitor cells (MPPs), which are Lin⁻, Sca-1⁺, c-kit(CD117)⁺, Ly6c⁻, FcεRI⁻, Thy1.1⁻, Flk-2⁺. The MPPs then either continue their development along the mast cell lineage by differentiating into common myeloid progenitors (CMPs) which are Lin⁻, Sca-1^{lo}, c-kit⁺, CD27⁺ and Flk-2⁻; or into common lymphoid progenitors (CLPs) which have not been found to be involved in mast cell precursor production. There is still debate about which lineage mast cells belong to from here, as CMPs branch into granulocyte/monocyte progenitors (GMPs) and megakaryocyte/erythrocyte progenitors (MEPs) during development within the bone marrow. GMPs can be found in either the spleen or bone marrow. In the bone marrow they are c-kit, CD34 and FcεRI positive and referred to as pre-BMPs that continue to develop into mast cell precursors(MCp) before leaving the bone marrow. They may also leave the bone marrow as GMPs and enter the spleen where they eventually differentiate into BMCPs. These BMCPs can then either differentiate into a basophil lineage or enter the blood and become MCp which are now identical to those MCp that formed in the bone marrow and are strictly on the mast cell path, maturing once they reach the peripheral tissue. In both mice and humans, mast cells have been found to originate from this common circulating MCp, with both MMC and CTMC arising from MCp. Two important markers of mast cell development are CD34 and FcεRI. Mast cells are CD34⁺ from when they are GMPs located within the bone marrow until they die, while FcεRI positivity is only seen as a signal of late committed MCp and until cell death.



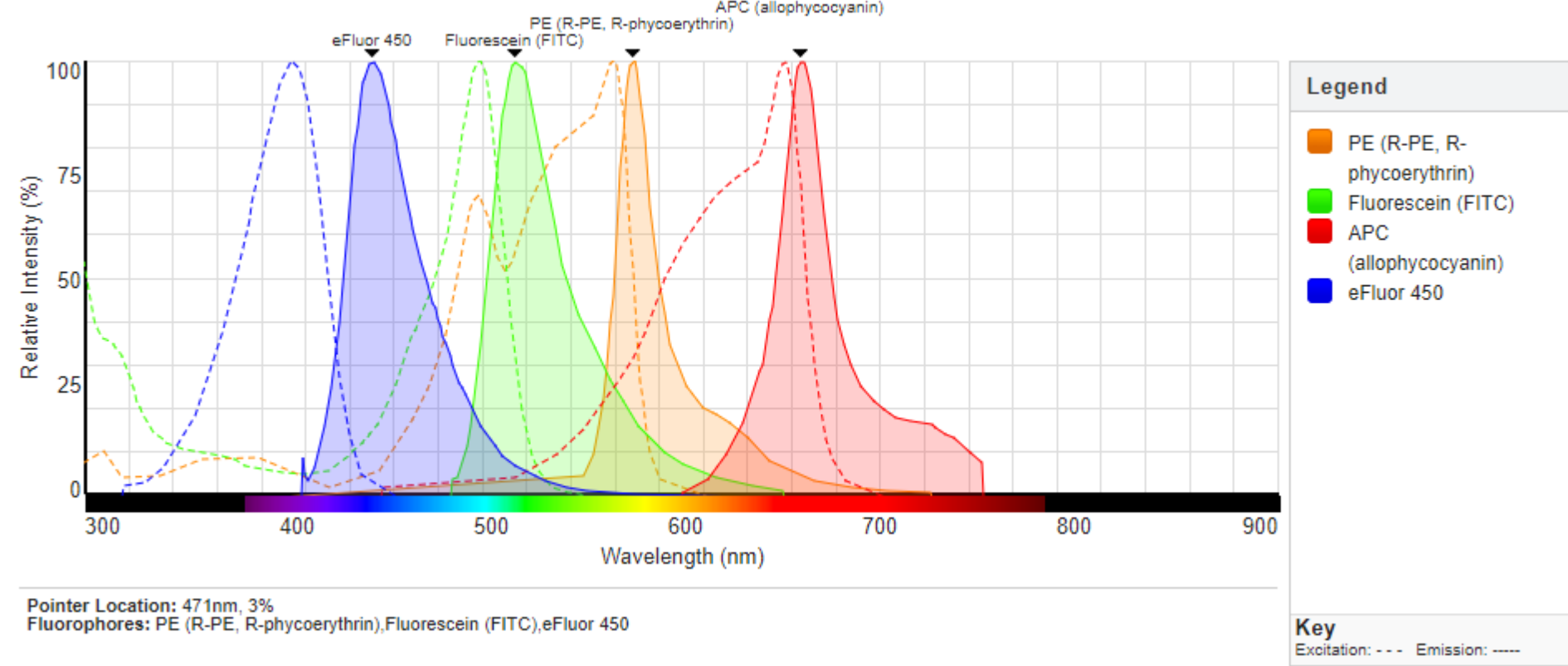
Antibodies, Cell Lines, and Controls

CD117 Antibody: Rat Anti-Mouse, Pig, Hamster, Fish, Guinea Pig CD117 (c-Kit) Monoclonal (2B8) PE-Conjugated Antibody (eBioscience cat no. 12-1171-82).

FcεRI Antibody: Goat anti-dog FcεRI (IgE) Polyclonal FITC conjugated antibody (BIO RAD cat no AHP946F).

CD90 Antibody: Rat anti-Dog CD90 (THY-1) monoclonal (YKIX337.217) APC conjugated antibody (eBioscience cat no. 17-5900-42).

Viability Dye: Ghost Dye Violet 450 (TONBO biosciences cat no. 10140-978)

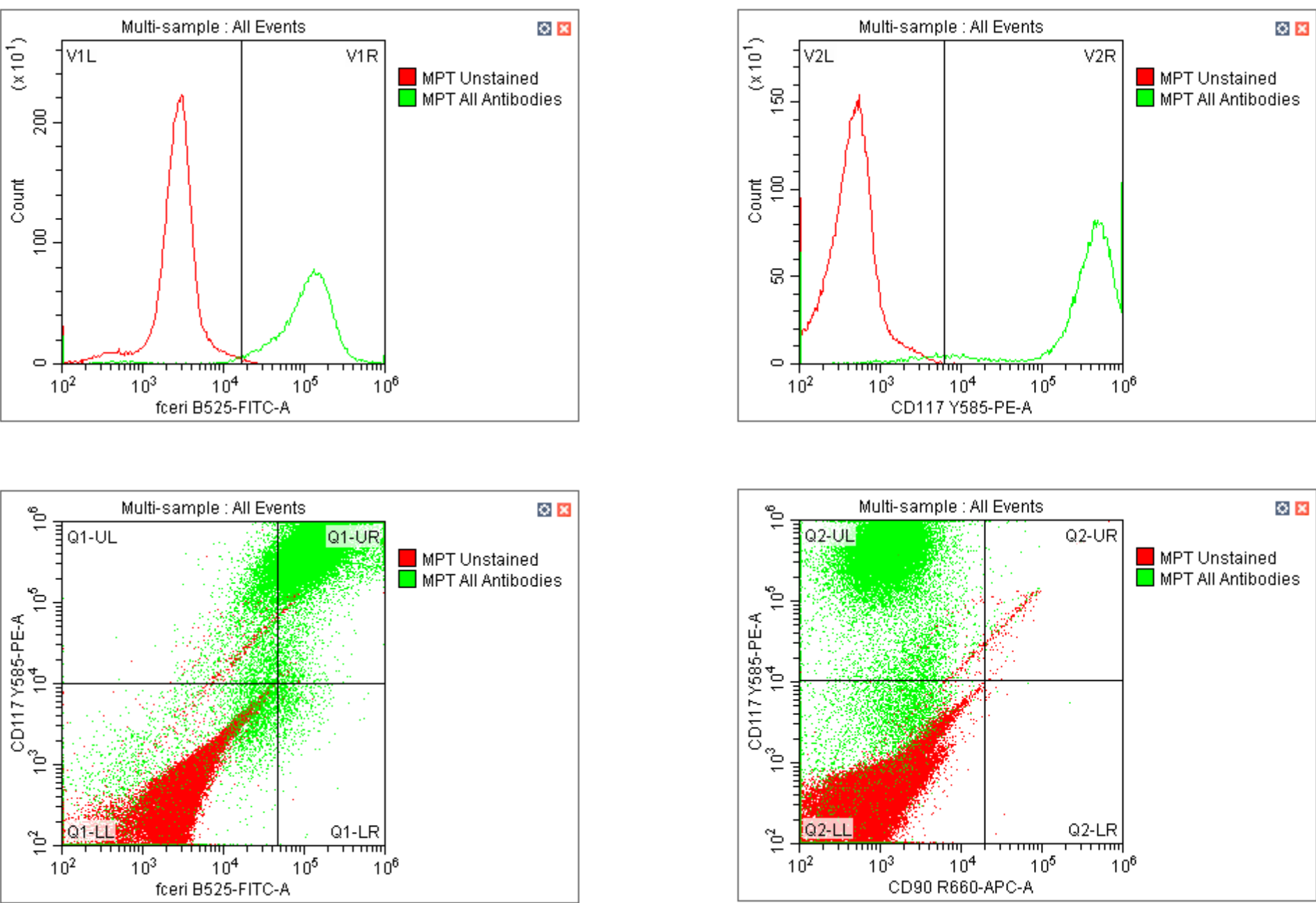


Cell Lines and Primary Cells:

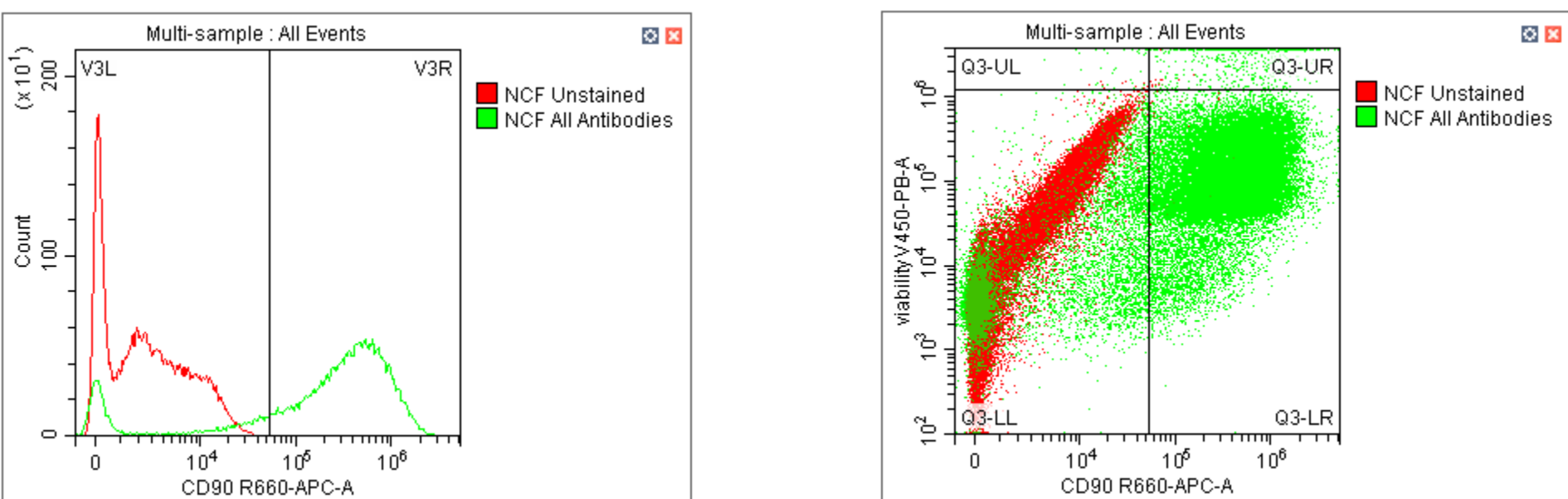
MPT-1: A mast cell tumor immortalized cell line used to positively identify CD117 and FcεRI antibody reactivity while also acting as a negative control for CD90 reactivity.

Normal Canine Fibroblasts (NCF): Primary cells used to identify positive CD90 reactivity and negative reactivity for CD117 and FcεRI.

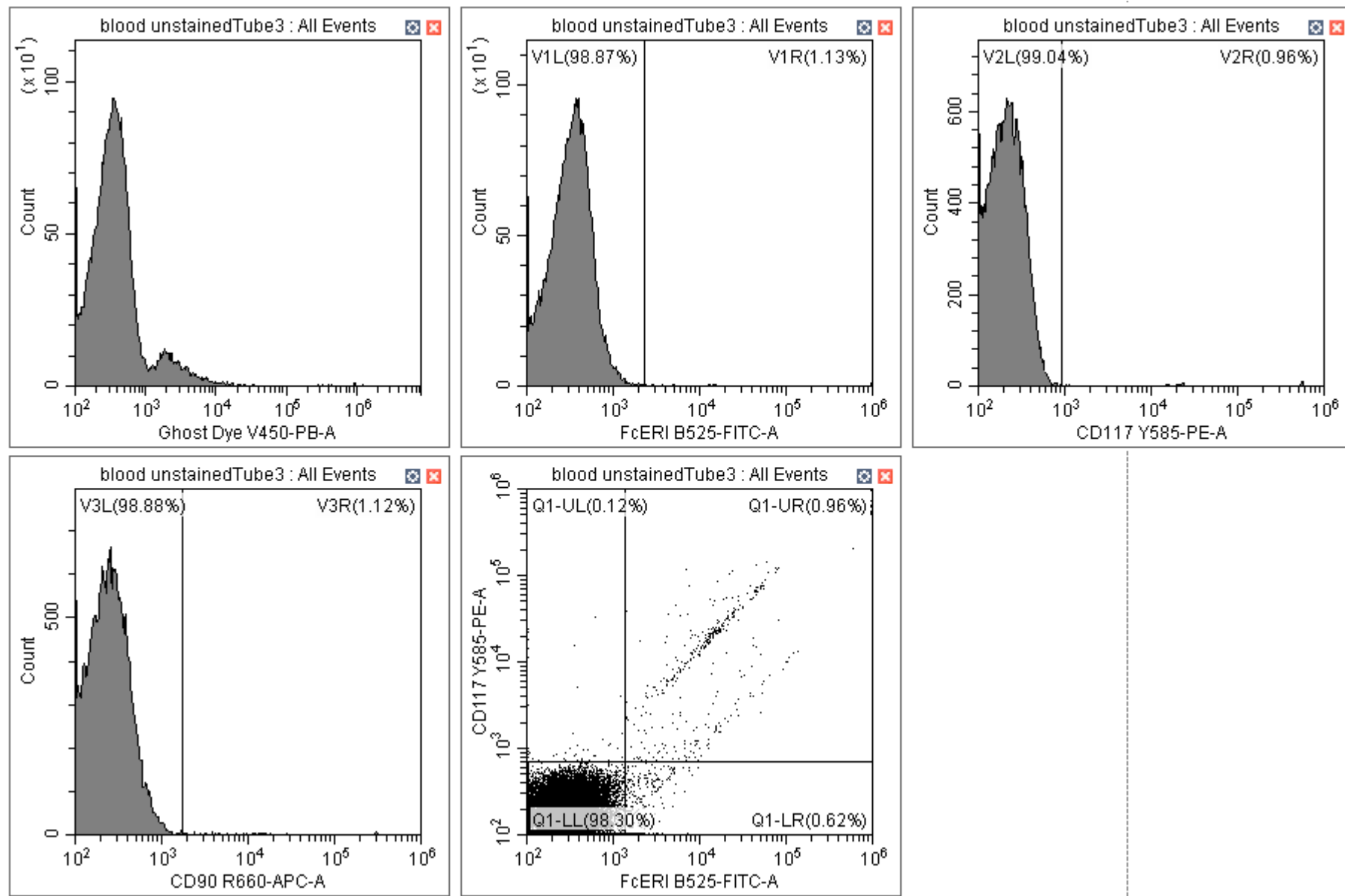
MPT-1 Canine Mast Cell Tumor Line CD117+ FcεRI+ CD90- Control



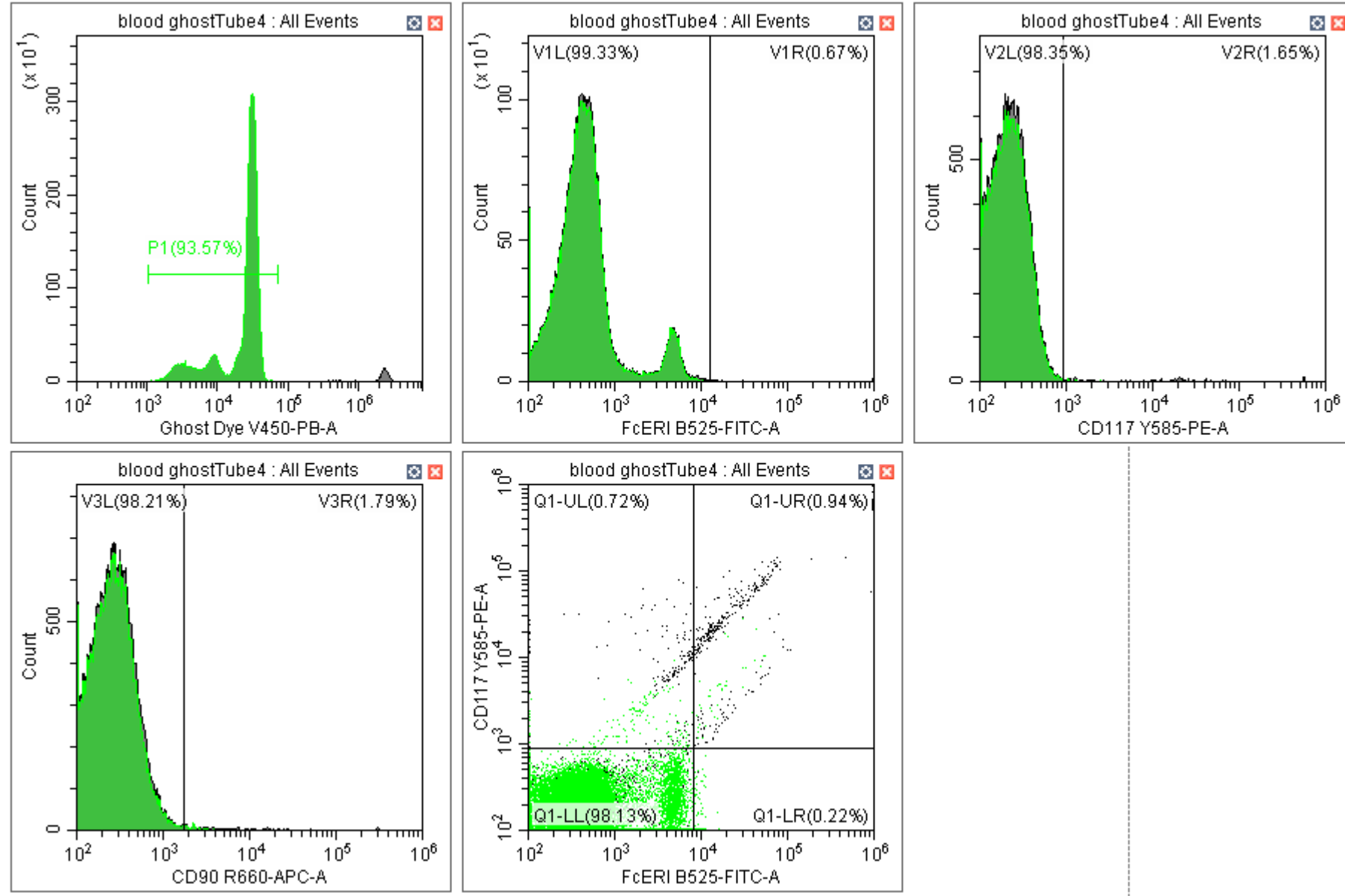
NCF Normal Canine Fibroblasts CD90+ CD117- FcεRI - Control



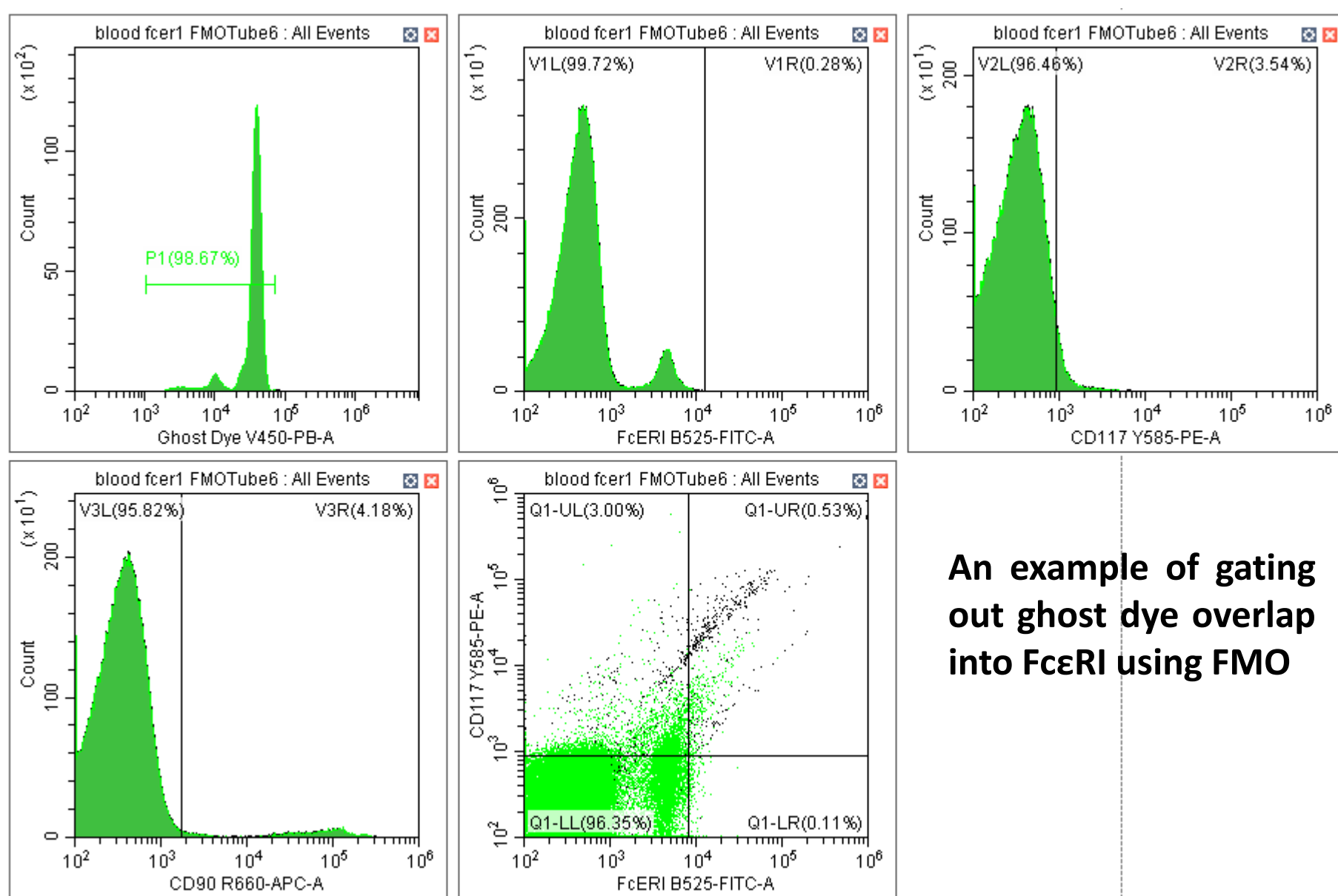
Gating Strategy



Gates are set on unstained samples first



Gates are then adjusted to focus on live cells using the ghost dye



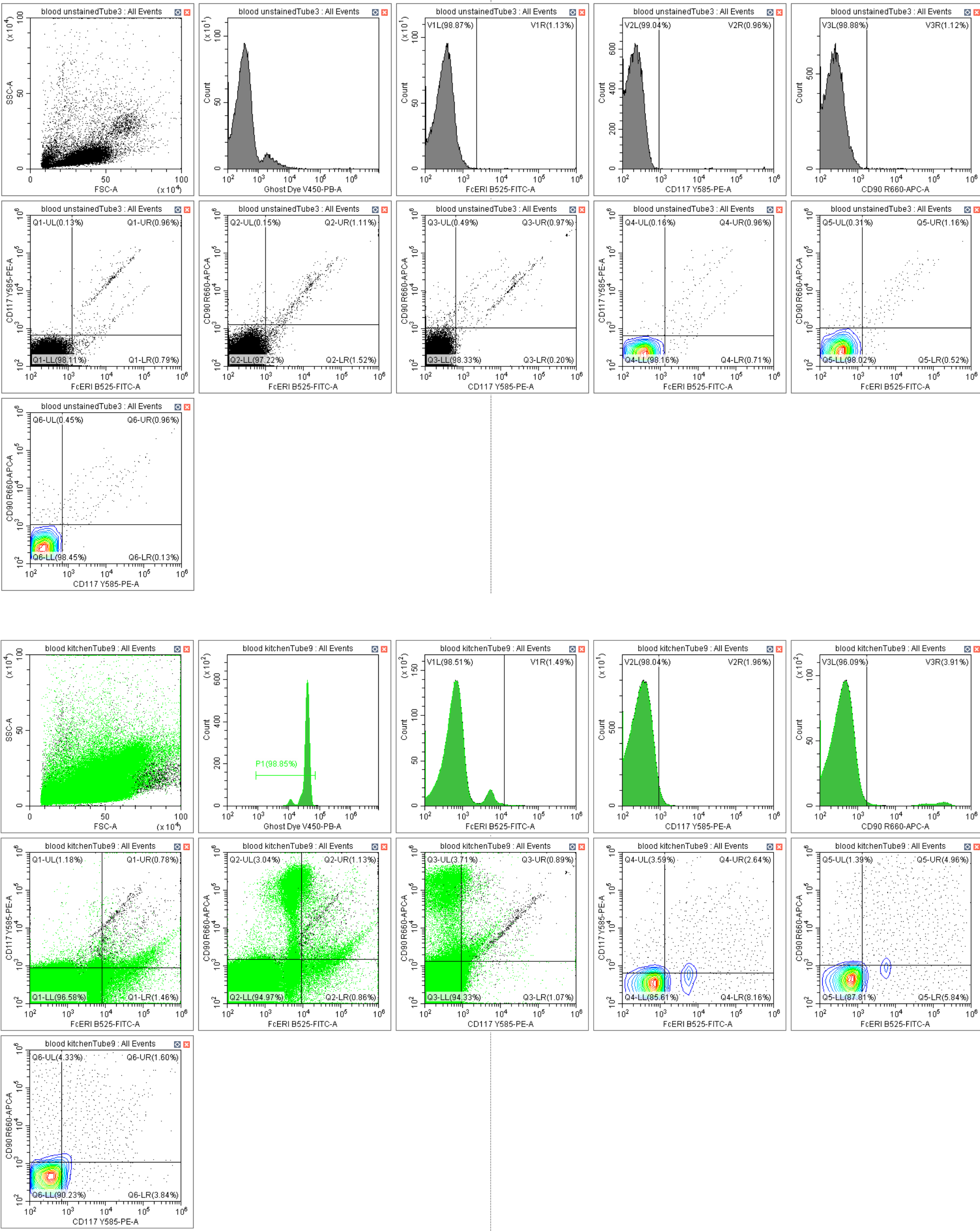
An example of gating out ghost dye overlap into FcεRI using FMO

Gates are finally adjusted using Fluorescence Minus One (FMO) and ready for analysis

What Is Fluorescence Minus One (FMO)?

Sample	PE	FITC	APC	NUV 450
Unstained	-	-	-	-
50:50 heat killed Ghost dye	-	-	-	+
CD117 FMO	-	+	+	+
FcεRI FMO	+	-	+	+
CD90 FMO	+	+	-	+
Ghost dye FMO	+	+	+	-
Kitchen Sink	+	+	+	+

Results



Conclusions

While we have yet to discern the CD117⁺ FcεRI⁺ CD90⁻ population from the general PBMC population, we remain confident that it is present. With the overall goal being to sequence this population, we believe that sorting of the population followed by single cell sequencing will allow for transcriptomic differentiation of the mast cell progenitor population.

Future Directions

Using both compensation and FMO we will be moving forward with flow analysis on the cell sorting machine, followed by sorting and sequencing of the CD117⁺ FcεRI⁺ CD90⁻ population. From this point we believe we will be able to distinguish a mast cell progenitor population that can be used as a normal for sequencing against mast cell tumor and with this data acquire genetic targets for the patient specific treatment of canine mastocytoma

References

Dahlin, J.S. and J. Hallgren, *Mast cell progenitors: origin, development and migration to tissues*. Mol Immunol, 2015. 63(1): p. 9-17.

Acknowledgements

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